Ziteng Song

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Education

Cornell University **Expected May 2027**
B.S. in Computer Science; Cumulative GPA: 4.00 *Ithaca, NY*

Technical Skills

Languages: Python, C++, Java, TypeScript, GLSL
Graphics & Vision: OpenCV, COLMAP, gsplat, WebGL 2, Three.js
ML & Tools: PyTorch, NumPy, SciPy, Pandas, Git

Experience

Cornell University — Imaging and HCI **Jan 2026 – Present**
Undergraduate Researcher; Advisor: Prof. Abe Davis *Ithaca, NY*

- Developed a pipeline to recover shared complex 3D vibration modes from unsynchronized multi-view videos using COLMAP, PyTorch, and gsplat.
- Separated static Gaussian geometry from frequency-dependent motion fields, accounting for cross-view phase and amplitude differences without frame-by-frame synchronization.
- Built an interactive Viser viewer to inspect recovered 3D vibration modes with adjustable amplitude and phase.

Wanka Huanju Culture Media Co., Ltd **May 2025 – Aug 2025**
Machine Learning Intern *Beijing, China*

- Built a Dify/VikingDB question-answering workflow connected to a WeChat public account, used internally for employee exam-question lookup.
- Built a preprocessing pipeline converting approximately 562,000 daily ad records into 181,040 observations across 496 product-placement time series.
- Trained and tuned a PyTorch Temporal Fusion Transformer, reducing MAE by 12.45% (79,180 to 69,324) versus a last-value baseline on a 14-day chronological validation horizon.

Teaching Assistant | Cornell University, CS 4620: Introduction to Computer Graphics **Fall 2026**

Projects

Physically Based C++ Renderer | *C++* | Report **Jan 2026 – May 2026**

- Extended Nori's educational C++ framework with BVH acceleration, MIS path tracing, dielectric/microfacet materials, and heterogeneous volumetric rendering with delta tracking.
- Rendered a 4000×3000 image at 8192 samples per pixel, combining textured geometry, environment lighting, and heterogeneous cloud volumes.
- Passed three volumetric sampling chi-square tests at $\alpha = 0.01$ (up to 200,000 samples/test); evaluated 10 image-validation cases, including five Mitsuba comparisons.

GPU Fluid Simulation & Volumetric Rendering | *TypeScript, GLSL* **Nov 2025 – Dec 2025**

- Developed a WebGL 2 fluid simulation and volumetric water renderer, achieving approximately 120 FPS with 40,000 particles on an NVIDIA RTX 4070 Laptop GPU.
- Implemented spatial hashing and bitonic sorting for GPU neighbor search; stored particle state in floating-point textures with ping-pong updates to avoid per-frame CPU readback.
- Resolved stale-density dependencies with an explicit density pass and bounded physics substeps; implemented volumetric ray marching, SDF collisions, and shoreline foam.

Gnome More Shrooms | *C++, CUGL* **Jan 2026 – May 2026**

- Served as the lead programmer and core gameplay designer on a 10-person mobile game team; earned Rookie Awards 2026 Game of the Year – Runner Up (Mobile).
- Implemented core gameplay mechanics and led architecture decisions, task planning, and integration across gameplay, multiplayer, and UI.
- Added frame-time profiling and draw-call counters; improved cache reuse, visibility culling, and pause/resume timing, largely eliminating sudden drops below 60 FPS during gameplay testing.